[例] $\frac{\partial}{\partial \bar{x}} \|x\|^2 = x$, $\frac{\partial}{\partial x} \|x\|^2 = \bar{x}$

$\frac{\partial}{\partial \bar{x}} \langle x, y \rangle = y$, $\frac{\partial}{\partial \bar{x}} \langle y, x \rangle = 0$

$x^H y = \sum \bar{x}_n \cdot y_n$

$\frac{\partial}{\partial \bar{x}} \mathrm{Re} \langle x, y \rangle = \frac{1}{2} y$

$\frac{\partial}{\partial \bar{x}} \frac{1}{2} (x^H y + y^H x)$

Least Square $f(x) = \|y - Ax\|_2^2$

1) $x \in \mathbb{R}^N$, $A \in \mathbb{R}^{M \times N}$ $\nabla f = 2A^T(Ax - y)$

2) $x \in \mathbb{C}^N$, $A \in \mathbb{C}^{M \times N}$ $\nabla f = 2A^H(Ax - y)$

$\frac{\partial}{\partial \bar{x}} f = A^H(Ax - y)$ $\frac{\partial}{\partial x} f = A^T(A^* x^* - y^*)$

Pf: ① 拆开 f

$$f(x) = (y - Ax)^H (y - Ax)$$
$$= y^H y - y^H A x - x^H A^H y + x^H A^H A x$$
$$\frac{\partial}{\partial \bar{x}} f = -A^H y + A^H A x = A^H (Ax - y)$$
$$\nabla f = 2 \frac{\partial}{\partial \bar{x}} f = 2A^H (Ax - y)$$

② Chain Rule

Let $r = y - Ax$, $\bar{r} = \bar{y} - \bar{A}\bar{x}$

$f = \|r\|^2 = r^H r$

$$\frac{\partial}{\partial \bar{x}} f = \underbrace{\left(\frac{\partial r}{\partial \bar{x}}\right)^T}_{0} \cdot \frac{\partial f}{\partial r} + \underbrace{\left(\frac{\partial \bar{r}}{\partial \bar{x}}\right)^T}_{-\bar{A}} \cdot \underbrace{\frac{\partial f}{\partial \bar{r}}}_{r}$$

$$= -A^H r = -A^H (y - Ax)$$
$$\nabla f = 2 \frac{\partial}{\partial \bar{x}} f = A^H (Ax - y)$$

3) $x \in \mathbb{R}^N$, $A \in \mathbb{C}^{M \times N}$ $\nabla_x f = 2\,\mathrm{Re}\{A^H(Ax - y)\}$

Pf: $f(x) = y^H y - y^H A x - x^T A^H y + x^T A^H A x$

$$= y^H y - \underbrace{(y^H A - y^T A^*)}_{2\mathrm{Re}(y^H A)} x + x^T A^H A x$$

此时 $f: \mathbb{R}^N \to \mathbb{R}$，用实数导数

$\nabla f = [2\mathrm{Re}(y^H A)]^T + 2(x^T A^H A$

$$\nabla f = -[2\mathrm{Re}(y^H A)] + \nabla(x^H \underbrace{A^H A}_{\text{复}} x)$$

$$= -2\mathrm{Re}(A^H y) + 2\mathrm{Re}(A^H A)x$$

$$= 2\mathrm{Re}\{A^H(Ax - y)\}$$

复内积空间 $\langle x, y\rangle = x^H y \in \mathbb{C}$

$\|x-y\|^2 = \|x\|^2 + \|y\|^2 + 2\mathrm{Re}\langle x, y\rangle$

Analytic

$z = x + jy \qquad x = \mathrm{Re}\, z,\ y = \mathrm{Im}\, z$

$f(z) = u(x,y) + jv(x,y) \qquad u(x,y) = \mathrm{Re} f,\ v(x,y) = \mathrm{Im} f$

可导 <=> Cauchy - Riemannian eq.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},\quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Analytic Func 解析：邻域内可导

初等函数解析 e.g. z^2, e^z

e.g. $|z|^2$ 仅在 0 处可导，不解析

$\|y - Az\|_2^2$, $\|Z\|_F^2$ 不解析

Wirtinger Derivative

不可导时，可定义

$$\frac{\partial f}{\partial z} := \frac{1}{2}\left[\frac{\partial f}{\partial x} - j\frac{\partial f}{\partial y}\right] \qquad \frac{\partial f}{\partial z^*} := \frac{1}{2}\left[\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y}\right]$$

$\left(\frac{\partial f}{\partial z}\right)^* = \frac{\partial f^*}{\partial z^*}$ 当 $f: \mathbb{C} \to \mathbb{R}$, $\left(\frac{\partial f}{\partial z}\right)^* = \frac{\partial f}{\partial z^*}$

当解析时，Wirtinger 导数中

$\frac{\partial f}{\partial z}$ 即解析导数，$\frac{\partial f}{\partial z^*}$ 即为 0

Pf：代入 C-R eq. 即可

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left[\frac{\partial f}{\partial x} - j\frac{\partial f}{\partial y}\right] = \frac{1}{2}\left[\frac{\partial u}{\partial x} + j\frac{\partial v}{\partial x} - j\left(\frac{\partial u}{\partial y} + j\frac{\partial v}{\partial y}\right)\right]$$

$$= \frac{1}{2}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + j\frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$\frac{\partial f}{\partial z^*} = \frac{1}{2}\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) + j\frac{1}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right) \overset{\text{CR eq}}{=} 0$$

Chain Rule

$$\frac{\partial}{\partial z} f(g(z)) = \frac{\partial f}{\partial g}\frac{\partial g}{\partial z} + \frac{\partial f}{\partial g^*}\frac{\partial g^*}{\partial z}$$

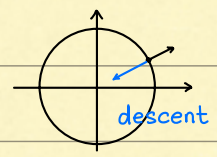
$$\frac{\partial}{\partial z^*} f(g(z)) = \frac{\partial f}{\partial g}\frac{\partial g}{\partial z^*} + \frac{\partial f}{\partial g^*}\frac{\partial g^*}{\partial z^*}$$

复变量优化

$f: \mathbb{C} \to \mathbb{R}$ Stationary： $\frac{\partial}{\partial x} f = 0$ / $\frac{\partial}{\partial \bar{x}} f = 0$

解析时，$-\nabla f$ 是下降方向

e.g. $f(z) = z^2$，则 $\frac{\partial}{\partial z} f = 2z$，$\frac{\partial}{\partial z^*} f = 0$



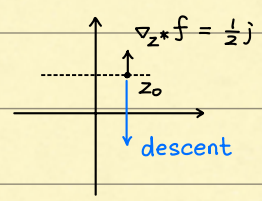
不解析时，$-\frac{\partial}{\partial x^*} f$ 是下降方向

Pf：$df = \frac{\partial}{\partial x} f\, dx + \frac{\partial}{\partial \bar{x}} f\, d\bar{x}$

取 $dx = -\alpha \frac{\partial}{\partial \bar{x}} f$，则 $df = -2\alpha \left|\frac{\partial}{\partial x} f\right|^2 < 0$

e.g. $f(z) = \mathrm{Im}(z)$，则 $\frac{\partial}{\partial z} f = -\frac{1}{2}j$，$\frac{\partial}{\partial z^*} f = \frac{1}{2}j$

必须让 $z^{k+1} = z^k - \alpha \cdot \frac{1}{2}j$ 才能往下走



$f: \mathbb{C} \to \mathbb{R}$

定义 $\nabla f = 2\frac{\partial}{\partial \bar{x}} f = \frac{\partial}{\partial \mathrm{Re}\, x} f + j\frac{\partial}{\partial \mathrm{Im}\, x} f$